

```
In [1]: import pandas as pd
import numpy as np
```

```
In [2]: df = pd.read_csv("D:\Dsbda practical\Facebook Metrics of Cosmetic Brand.csv")
```

```
In [3]: df
```

Out[3]:

	Unnamed: 0	Page total likes	Type	Category	Post Month	Post Weekday	Post Hour	Paid	Lifetime Post Total Reach	F
0	0	139441	Photo	2	12	4	3	0.0	2752	
1	1	139441	Status	2	12	3	10	0.0	10460	
2	2	139441	Photo	3	12	3	3	0.0	2413	
3	3	139441	Photo	2	12	2	10	1.0	50128	
4	4	139441	Photo	2	12	2	3	0.0	7244	
...	...	...	...	...	...	...	...	...	...	
495	495	85093	Photo	3	1	7	2	0.0	4684	
496	496	81370	Photo	2	1	5	8	0.0	3480	
497	497	81370	Photo	1	1	5	2	0.0	3778	
498	498	81370	Photo	3	1	4	11	0.0	4156	
499	499	81370	Photo	2	1	4	4	NaN	4188	

500 rows × 20 columns



```
In [6]: print("Original Dataset:")
print(df.head())
print("-" * 50)
```

Original Dataset:

	Unnamed: 0	Page total likes	Type	Category	Post Month	Post Weekday	\
0	0	139441	Photo	2	12	4	
1	1	139441	Status	2	12	3	
2	2	139441	Photo	3	12	3	
3	3	139441	Photo	2	12	2	
4	4	139441	Photo	2	12	2	

	Post Hour	Paid	Lifetime Post Total Reach	\
0	3	0.0	2752	
1	10	0.0	10460	
2	3	0.0	2413	
3	10	1.0	50128	
4	3	0.0	7244	

	Lifetime Post Total Impressions	Lifetime Engaged Users	\
0	5091	178	
1	19057	1457	
2	4373	177	
3	87991	2211	
4	13594	671	

	Lifetime Post Consumers	Lifetime Post Consumptions	\
0	109	159	
1	1361	1674	
2	113	154	
3	790	1119	
4	410	580	

	Lifetime Post Impressions by people who have liked your Page	\
0	3078	
1	11710	
2	2812	
3	61027	
4	6228	

	Lifetime Post reach by people who like your Page	\
0	1640	
1	6112	
2	1503	
3	32048	
4	3200	

	Lifetime People who have liked your Page and engaged with your post	\
0	119	
1	1108	
2	132	
3	1386	
4	396	

	comment	like	share	Total Interactions
0	4	79.0	17.0	100
1	5	130.0	29.0	164
2	0	66.0	14.0	80
3	58	1572.0	147.0	1777
4	19	325.0	49.0	393

```
In [7]: subset = df[["Type", "Post Hour", "Lifetime Post Total Reach"]]
```

```
print("Data Subset (Selected Columns):")
print(subset.head())
print("-" * 50)
```

Data Subset (Selected Columns):

	Type	Post Hour	Lifetime Post Total	Reach
0	Photo	3		2752
1	Status	10		10460
2	Photo	3		2413
3	Photo	10		50128
4	Photo	3		7244

```
In [8]: extra_data = pd.DataFrame({
        "Type": ["Photo", "Status", "Video", "Link"],
        "Content Score": [80, 65, 90, 70]
    })
```

```
In [9]: merged_data = pd.merge(df, extra_data, on="Type", how="left")

print("Merged Data:")
print(merged_data.head())
print("-" * 50)
```

Merged Data:

	Unnamed: 0	Page total likes	Type	Category	Post Month	Post Weekday	\
0	0	139441	Photo	2	12	4	
1	1	139441	Status	2	12	3	
2	2	139441	Photo	3	12	3	
3	3	139441	Photo	2	12	2	
4	4	139441	Photo	2	12	2	

	Post Hour	Paid	Lifetime Post Total Reach	\
0	3	0.0	2752	
1	10	0.0	10460	
2	3	0.0	2413	
3	10	1.0	50128	
4	3	0.0	7244	

	Lifetime Post Total Impressions	...	Lifetime Post Consumers	\
0	5091	...	109	
1	19057	...	1361	
2	4373	...	113	
3	87991	...	790	
4	13594	...	410	

	Lifetime Post Consumptions	\
0	159	
1	1674	
2	154	
3	1119	
4	580	

	Lifetime Post Impressions by people who have liked your Page	\
0	3078	
1	11710	
2	2812	
3	61027	
4	6228	

	Lifetime Post reach by people who like your Page	\
0	1640	
1	6112	
2	1503	
3	32048	
4	3200	

	Lifetime People who have liked your Page and engaged with your post	\
0	119	
1	1108	
2	132	
3	1386	
4	396	

	comment	like	share	Total Interactions	Content Score
0	4	79.0	17.0	100	80
1	5	130.0	29.0	164	65
2	0	66.0	14.0	80	80
3	58	1572.0	147.0	1777	80
4	19	325.0	49.0	393	80

[5 rows x 21 columns]

```
In [10]: sorted_data = df.sort_values(  
        by="Lifetime Post Total Reach",  
        ascending=False  
    )
```

```
In [11]: print("Sorted Data (by Lifetime Post Total Reach):")  
        print(sorted_data.head())  
        print("-" * 50)
```

Sorted Data (by Lifetime Post Total Reach):

	Unnamed: 0	Page total likes	Type	Category	Post Month	Post Weekday	\
244	244	130791	Photo	2	7	3	
464	464	92079	Photo	1	2	6	
463	463	92186	Photo	3	2	7	
277	277	126424	Video	1	6	2	
380	380	111620	Photo	1	4	7	

	Post Hour	Paid	Lifetime Post Total Reach	\
244	5	1.0	180480	
464	13	0.0	158208	
463	2	1.0	153536	
277	13	0.0	139008	
380	14	0.0	128064	

	Lifetime Post Total Impressions	Lifetime Engaged Users	\
244	319133	8072	
464	453213	2482	
463	497910	1713	
277	277100	1779	
380	251269	1539	

	Lifetime Post Consumers	Lifetime Post Consumptions	\
244	4010	6242	
464	2319	3412	
463	1633	2493	
277	1643	2356	
380	1408	2506	

	Lifetime Post Impressions by people who have liked your Page	\
244	108752	
464	81938	
463	8907	
277	107502	
380	84046	

	Lifetime Post reach by people who like your Page	\
244	51456	
464	20608	
463	5696	
277	38720	
380	32384	

	Lifetime People who have liked your Page and engaged with your post	\
244	3316	
464	1034	
463	593	
277	1008	
380	985	

	comment	like	share	Total Interactions
244	372	5172.0	790.0	6334
464	9	268.0	36.0	313
463	8	134.0	34.0	176
277	23	204.0	44.0	271
380	8	186.0	18.0	212

```
In [12]: transposed_data = df.head().T
```

```
print("Transposed Data:")
print(transposed_data)
print("-" * 50)
```

Transposed Data:

	0	1	2 \
Unnamed: 0	0	1	2
Page total likes	139441	139441	139441
Type	Photo	Status	Photo
Category	2	2	3
Post Month	12	12	12
Post Weekday	4	3	3
Post Hour	3	10	3
Paid	0.0	0.0	0.0
Lifetime Post Total Reach	2752	10460	2413
Lifetime Post Total Impressions	5091	19057	4373
Lifetime Engaged Users	178	1457	177
Lifetime Post Consumers	109	1361	113
Lifetime Post Consumptions	159	1674	154
Lifetime Post Impressions by people who have li...	3078	11710	2812
Lifetime Post reach by people who like your Page	1640	6112	1503
Lifetime People who have liked your Page and en...	119	1108	132
comment	4	5	0
like	79.0	130.0	66.0
share	17.0	29.0	14.0
Total Interactions	100	164	80

	3	4
Unnamed: 0	3	4
Page total likes	139441	139441
Type	Photo	Photo
Category	2	2
Post Month	12	12
Post Weekday	2	2
Post Hour	10	3
Paid	1.0	0.0
Lifetime Post Total Reach	50128	7244
Lifetime Post Total Impressions	87991	13594
Lifetime Engaged Users	2211	671
Lifetime Post Consumers	790	410
Lifetime Post Consumptions	1119	580
Lifetime Post Impressions by people who have li...	61027	6228
Lifetime Post reach by people who like your Page	32048	3200
Lifetime People who have liked your Page and en...	1386	396
comment	58	19
like	1572.0	325.0
share	147.0	49.0
Total Interactions	1777	393

```
In [13]: print("Shape of Dataset:")
print(df.shape)
print("-" * 50)
```

Shape of Dataset:
(500, 20)

```
In [14]: reach_array = np.array(df["Lifetime Post Total Reach"].head(6))
reshaped_array = reach_array.reshape(2, 3)
```

```
In [15]: print("Reshaped Array (2x3):")
print(reshaped_array)
print("-" * 50)
```

```
Reshaped Array (2x3):
[[ 2752 10460 2413]
 [50128 7244 10472]]
-----
```

```
In [16]: pivot_data = df.pivot_table(
    values="Lifetime Post Total Reach",
    index="Type",
    columns="Post Month",
    aggfunc="mean"
)
```

```
In [17]: print("Pivot Table:")
print(pivot_data)
print("-" * 50)

print("All operations completed successfully.")
```

```
Pivot Table:
Post Month      1      2      3      4      5  \
Type
Link      23906.500000  9356.0  5179.666667  46821.333333  12540.000000
Photo      10155.809524  24214.8  14064.937500  16834.434783  10285.212121
Status              NaN      NaN  9120.000000  17912.000000  14405.333333
Video              NaN      NaN      NaN      NaN      NaN

Post Month      6      7      8      9  \
Type
Link              NaN      NaN      NaN      NaN
Photo      14350.195652  20391.863636  11956.833333  6363.645161
Status      10336.000000  14074.285714  13022.000000  13673.000000
Video      139008.000000  21872.000000      NaN  30624.000000

Post Month      10      11      12
Type
Link      25757.333333  21176.000000  6001.333333
Photo      8143.367347  11527.184211  8509.870968
Status      14448.500000  8158.666667  12688.833333
Video              NaN  43576.000000  36208.000000
-----
All operations completed successfully.
```

```
In [ ]:
```